

The rumen is not a “black box”

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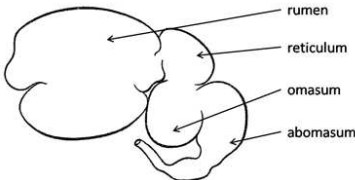
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Increasing efficiency in the feeding systems is an important contributing factor to farm profitability. Feed efficiency is a complex function of nutrient intake in relation to growth-performance potential. Because roughly 80% of the total tract digestible organic matter intake is fermented in the rumen, an understanding of the principles of fermentation processes and the limits of the rumen function are key to optimizing feed efficiency.

The rumen

The rumen is the first (and larger) compartment of the four compartments that compose the stomach of ruminants. The rumen acts as a fermentation chamber, thus two major conditions must be present to carry out the fermentation process: 1) the rumen must be anatomically and physiologically developed to perform its fermentation processes (Table 1), and 2) fermentive adaptations within the rumen needs to be developed. This is mediated by introduction of microorganism from feed, environment and other animals to the rumen. Products of fermentation of soluble carbohydrates (i.e. starch) promote the papillae development, while intake of forages promote muscular activity (and promote the increases of rumen size). The normal characteristics of a functional rumen are described in Table 2.

Table 1. Changes in relative volume of the four compartments of the rumen system

	Compartment	Newborn	2-3 Months	Adult
	Reticulum	5 %	5 %	5 %
	Rumen	25 %	65 %	80 %
	Omasum	10 %	10 %	7-8 %
	Abomasum	60 %	20 %	7-8 %

Adapted from Van Saun (2006)

Table 2. Environment characteristics of a functional rumen

Item	Characteristic
pH	5.5 -7.2
Osmolarity	~ 280 mOsm
Potential Redox	-250 to -450 mV
Temperature	38 – 41°C
Gas phase	Anaerobic, CO ₂ (65%), CH ₄ (27%), N ₂ (7%), O ₂ (0.6%)
Solid phase	Fibrous mat
Liquid phase	Volatile Fatty Acids (VFA), Ammonia, Minerals, Soluble protein
VFA's	Acetate, Propionate, Butyrate
Bacteria	50-60% total microbial mass
Protozoos	30-40% total microbial mass
Fungus	8% total microbial mass

Adapted from Yokohama and Johnson (1988) and from Owens and Goetsch (1988)

Ruminal fermentation

The diverse ruminal microbiota forms a key link between ruminants and their diet through the production of metabolic wastes (volatile fatty acids and gases), synthesis of microbial biomass (constituents of microbial growth such as protein and vitamins), and detoxification of otherwise noxious feed components. Although net microbial biomass synthesis and related VFA production are a rather predictable function of digestible organic matter intake (Figure 1), factors that affect the efficiency of the process are complex, and the subject of intense scientific investigation.

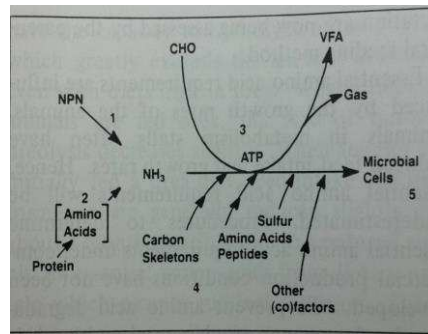


Figure 1 Flow diagram of the major reactions for carbohydrate fermentation, nitrogen catabolism and cell biosynthesis (Owens and Zinn, 1988)

Microbial protein

Microbes contain approximately 50% true protein that has an intestinal digestibility of about 80%. The protein is considered high quality as it is a good source of both methionine and lysine. Microbial protein may account for greater than 60% of protein supply to the small intestine. However, the efficiency microbial protein synthesis (net flow of microbial true protein to the small intestine) can be quite variable, ranging from 75 to 150 g microbial true protein per kilogram of organic matter fermented. Thus, the interest in better understanding feed factors that impact efficiency. The rumen microbial biomass is a consortium of bacteria, ciliate protozoa, flagellate protozoa, phycomycete fungi, amoebae and bacteriophages. The bacteria accounts for about 50% of the total microbial mass (10^9 - 10^{11} cells / g of ruminal contents), protozoa represents 40% (10^5 - 10^6 cells/g of rumen contents) and fungi about 8% of the microbial biomass (Yokohama and Johnson, 1988).

Microbial efficiency and microbial yield

Microbial efficiency can also be defined as proportion of substrate energy fixed into cells. Efficiency is affected by nutrient deficiencies (ie. insufficient ruminal ammonia N), and proportion of protozoa (Fig 2), as well as ruminal pH (lower pH lower efficiency) and ruminal passage rate (greater passage rate of microbes from the rumen, greater efficiency).

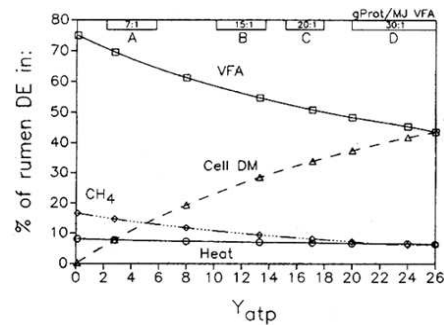


Figure 2. Relationship between the production of microbial cell and VFA and methane in fermentative digestion of ruminants (Leng, 1993).

Microbial protein yield: Accordingly to the NRC (1996) the microbial protein yield can be estimated by applying the equation: Microbial protein yield, g = $0.13 \times \text{TDN}_{\text{kg}} \times \text{eNDF factor}$, where the coefficient 0.13 means the microbial efficiency, TDN is the total digestible nutrients, and eNDF factor represent the amount of effective neutral detergent fiber in diet, if $\text{eNDF} \geq 20$, then, eNDF factor = 1.

This equation is functional as long as the ration contains $\geq 64\%$ TDN and $\geq 20\%$ eNDF (effective neutral detergent fiber), otherwise, the following adjustments must be made:

If $\text{TDN} < 64$, a reduction of 0.0029 of the 0.13 value for each percent unit below of 64, for example if $\text{TDN} = 60$, then the constant value for microbial efficiency = 0.118 ($0.13 - (0.0029 \times 4)$)

On another hand, if $\text{eNDF} < 20$, a reduction of 0.022 for each percent unit below 20 must be subtracted to 1, example: eNDF in ration = 18, thus eNDF factor = $1 - (0.022 \times 2) = 0.956$. In this sense, a steer that fed 7 kg DM basis of a ration that contains 66%TDN and 21%NDF, the estimated microbial yield is 574 g of microbials/day, the same animal fed the same quantity of feed but the rations contains 62%TDN and 18% eNDF, then, the microbial protein yield resulted in 515 g represent a reduction of 10.2%. Accordingly, when diet protein concentration is non-limiting, MP synthesis is affected by the level of disposable energy and by the concentration of effective NDF in diet (as it relates to ruminal pH and rate of passage). In all, the amount of ruminally degradable N required to optimize microbial growth is largely dependent on digestibility of the diet. For example, a steam-flaked corn-based diet will require more ruminally available N than a dry rolled corn-based diet.

Resistance of feed proteins to microbial degradation may limit microbial protein synthesis. Whereas, ruminal degradation of high-value proteins (such soybean meal) results in inefficient utilization of intact feed protein. With high-energy finishing diets, supplementation with NPN (urea) can enhance growth–performance to a greater extent than supplementation with intact protein (Zinn et al., 2003). May et al. (2014) evaluated supplemental urea levels of 0.1, 0.70 and 1.25% in a steam-flaked corn based finishing diet (90:10 concentrate:forage ratio), the urea fermentation potential (UFP) corresponded to 1.2, 0.6 and 0, respectively. As urea increased on diet, microbial N flows, digestibility of OM and daily weight gain and dietary energetic efficiency increased (Table 3). When the rate of carbohydrate fermentation exceeds supply of N required for growth, efficiency of microbial protein synthesis decreases (Nocek and Russell, 1988). However this is not to be confused with the popular notion that rate of soluble feed N release to the rumen be in synchrony with carbohydrate fermentation. Numerous

studies have proved the concept indefensible. Providing adequate ruminal available N, irregardless of the rate at which it is degraded or solubilized within the rumen, is the relevant factor affecting microbial protein synthesis. Irregardless of source (NPN or intact protein), microbial protein synthesis is maximal when degradable intake protein is roughly 10% of digestible organic matter intake. This effect is due to N recycling to the rumen (Stern et al., 2006). However, in a few recent studies evaluating urea supplementation of growing finishing diets (López-Soto et al., 2014a,b), combining urea with a “slow release” urea product enhanced microbial nitrogen flow to the small intestine, digestible energy, and growth performance (Tables 4 & 5).

Synthesis of microbial protein is improved by varying the source and degradability of energy constituents incorporated into the diet (Kaswari, 2004). An excess of the soluble carbohydrates promote a lower ruminal pH over an extended period, affecting the growth of cellulolytic bacteria decreasing the rate of fiber digestion. Other factor that affect microbial yield is the passage rate [defined as the rate (% / h) in which matter leaves the rumen]. At greater passage rates, microbial efficiencies are increased due to a reduction in maintenance energy costs (Fellner, 2009). Passage rate increases with decreased eNDF content of feeds. Therefore it can be expected that, when cattle fed finishing diets, the microbial yield is greater, but microbial efficiency may be lower than when cattle are fed a growing diets (Figure 3), due to greater extent of ruminal organic matter digestion.

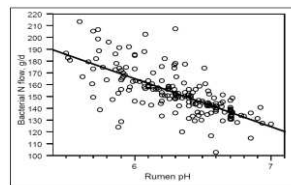


Figure 3. Taken from Stern et al. (2006)

Volatile fatty Acids

The molar proportions of the rumen volatile fatty acids (VFA, especially acetic and propionic) are related to the energetic efficiency of weight gain. VFA's are produced mainly during the digestion of carbohydrates and fermentation of sugars but also during the digestion of amino acids. The three main volatile fatty acids that are produced by rumen fermentation are: acetate, propionate and butyrate which have 2, 3 and 4 carbons, respectively. Energy derived from ruminal VFA production may account for up to 80% of the animals maintenance energy requirements (Bergman, 1990). Ruminal VFA production is closely related to ruminal pH, which (as mentioned previously) is an important regulator of microbial yield. A practical way to approach the energy efficiency of VFA production is the acetate:propionate molar ratio. Propionate is utilized more efficiency than acetate and butyrate (Figure 4). The proportions in which the VFA's are produced in rumen are affected by physical and chemical composition of diets. At greater acetate:propionate ratio, reflects greater methane (CH₄) energy loss, and decreased efficiency of VFA energy utilization. Factors that affect ruminal VFA molar ratios include forage level, particle size, and the proportion of rapidly fermentable carbohydrates. Increasing forage level increases acetate. Alternatively, increasing concentrates in rations increases the proportion of propionate (Figure 5).

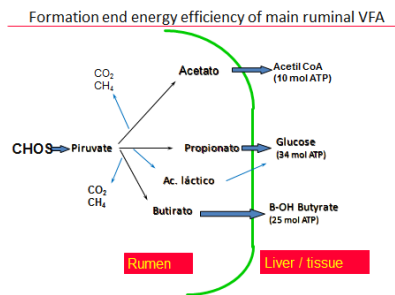


Figure 4. Formation and energy efficiency in ruminant tissues of main ruminal VFA

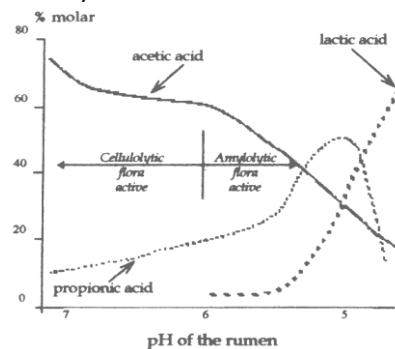


Figure 5. Relationship between ruminal pH and proportions of acetic, propionic, and lactic acid produced (Kaufmann et al., 1980)

Manipulation of ruminal fermentation to increase net energy of diet

Ruminal organic matter digestion as well as VFA molar proportions can be manipulated by dietary modifications and/or addition of feed additives (Hernández-Sarabia et al., 2012). With respect to feed additives, ionophores have been used for decades in ruminant diets to improve energetic efficiency (Duffield et al., 2012). The ionophore induced improvement in animal performance is explained in terms of their selective action on the gram positive bacteria in the rumen. Ionophores may enhance the capture of feed carbon as propionate rather than methane resulting in the improvement in feed energy utilization. Ionophores may inhibit lactic acid production, and reduce amino acid deamination resulting enhanced N use (Russell and Strobel, 1988; Zinn et al., 1994). However, the magnitude of the response to ionophores supplementation has been variable, ranging from nil or even negative (Stock et al., 1990; Zinn and Borques, 1993; Depenbusch et al., 2008) to greater than 18% (Bartley et al., 1979). The basis for the variable response is not clear. However, diet energy density has been implicated (Zinn et al., 1994; Barreras et al., 2013). In a meta-analysis performed by Duffield et al. (2012), they noted that, in the last 40 years, the impact of monensin on feed efficiency has decreased from 8.1 to 3.5% (Table 6). This change may be explained partially by increases in diet energy density in present feedlot feeding systems (Zinn et al., 2011; Duffield et al., 2012) and by the reduction of fiber contents in diets, factors that of themselves reduce acetate:propionate molar ratio.

In some countries the use of antibiotics, such as ionophores, as feed additives is prohibited. Although the safety of these additives has been verified through exhaustive

evaluations, there remains some concern among consumers, and hence considerable effort is underway to advance the use of more “organic” alternatives. The latter perception has generated interest in the search for generally-recognized-as-safe alternatives. Among these, several compounds as organic acids, yeast, enzymes, and plant extracts are becoming more prevalent.

The organic acids are another class of feed additives used to enhance ruminal fermentation. Whereas ionophores inhibit certain bacteria, organic acids (such malic, fumaric, and aspartic acids) exert their influence by stimulating specific microbes (Nisbet and Martin, 1993). Irrespective of their mode of action, organic acids supplementation may increase propionate, lower lactic acid and methane by rumen microbes, and improve average daily gain and feed efficiency feedlot cattle (Sanson and Stallcup, 1988) and milk production (Kung et al., 1982; Table 7; Martínez-Gonzales et al., 2014).

Ruminal bacteria do not grow when the pH level is below 6.0. When the level is between 6.5 and 6.0, bacterial growth declines at the rate of 14% per hour for each 0.1 decrease pH. Optimal growth is achieved when pH is above 6.5. A deficiency of fibrolytic enzymes that can occur when ruminal pH is low may be partially overcome by introduction direct-fed microbials or by enzyme supplementation. Direct-fed microbials, such as yeast cultures (YC) have shown promise, although responses have not been consistent. In a few studies, YC supplementation enhanced NDF digestion (Table 8; López-Soto et al., 2013) or ruminal microbial protein synthesis (Newbold *et al.*, 1996; Mukhtar *et al.*, 2010). However, in other cases no beneficial effects were observed in characteristics of digestion with YC supplementation (Beauchemin et al., 2003; McGinn et al., 2004). Apparently, the efficacy of YC supplementation on growing-finishing diets depends on the dose of YC administered (Domínguez-Vara *et al.*, 2009) and the diet composition (forage:concentrate ratio; Galip, 2006). As a result of the rumen environment conditions in cattle fed high concentrate diets (Fernando et al., 2010), there is now a general consensus about the more pronounced and clear effects of live yeast with concentrated diets than with fibrous diets (Denev, 2007; Khalid et al. 2011).

Supplementation of high-energy diets finishing diets with fibrolytic enzymes has enhanced fiber digestibility both in vivo and in vitro. [Krause et al. \(1998\)](#) reported increased total tract digestibility of acid detergent fiber when feedlot cattle fed barley-based diets were supplemented with fibrolytic enzymes. Lewis et al. (1996) also reported increased total tract digestibility of DM and NDF when feeding fibrolytic enzymes to beef steers. Increases on NDF turnover have been reported when cattle fed medium to high-energy diets supplemented with fibrolytic enzymes (Ware and Zinn, 2005; López-Soto et al., 2006). Because the rumen has an upper limit on its physical capacity, the rate in which the ruminal fiber are digested directly affect DM intake in cattle. As mentioned previously, in high-energy diets, cellulolytic bacteria are compromised and even small changes on ruminal NDF digestion can have significant impacts on cattle performance. For example, with a 220 kg calf fed a growing diet containing 21% NDF, a 20% reduction in ruminal fiber digestion is expected to depress ADG by 10% (1.15 vs. 1.28 kg/d). With a 630 kg lactating cow fed a diet containing 30% NDF (80% eNDF), decreasing ruminal NDF digestion from 45% to 35% would drop maximum DM intake from 25 to 21 kg/d, resulting in a 21% decrease in milk yield (37 vs. 47 kg/d; Zinn et al., 2007). Increases in rate of weight gain as result of supplementation with YC or fibrolytic enzymes could be

explained mainly by changes in digestibility of fiber fractions (Khalid et al. 2011). These effects generally permit greater intakes and daily gains without effects on feed efficiencies or on the NE value of the diet (Table 9).

Recently, the use of phytogetic compounds has shown promise as feed additives in ruminants diets. Some of the naturally occurring plant components like polyphenols (tannins), detergents (saponins) and isoquinoline alkaloids (i.e., sanguinarine, chelerythrine, allocryptopin) have the potential to modulate rumen fermentation. The characteristics of tannins and its effect on ruminal fermentation and productivity have been reviewed extensively previously (Makkar, 2003; Frutos et al., 2004). The main *in vitro* effects of tannins are decreases in rate of feed protein digestion, and reductions in ruminal protozoal counts (Makkar et al., 1998). However, results on animal productivity have not been consistent. Rivera-Mendez et al. (2014) observed that whereas supplementation with 0.6% tannin increased feed intake and corresponding daily weight gain, it did not enhance dietary energetic efficiency (net energy value of the diet; Table 10). Saponins may facilitate a greater organic matter digestion and microbial mass production (Hu et al., 2005). Saponins may reduce methane energy loss (Jayanegara et al., 2014). Goats fed with 3 g / day of saponins had greater weight gains and feed efficiency in one study (Hu et al., 2006), but in another study, no effects on growth performance or feed efficiency were observed (Nasri et al., 2011). Other phytogetic compounds, such as isoquinoline alkaloids (ISQ-ALK) are known to have antimicrobial (Newton et al., 2002) and anti-inflammatory (Tanaka et al., 1993) properties. In broilers and pigs ISQ-ALK supplementation had positive effects on health and performance (Vieira et al., 2007; Zhou et al., 2013). Supplementation with ISQ-ALK decreased breakdown of feed protein (Drsata et al., 1996), the basis for decreased concentration of ammonia *in vitro* (Smink and Van der Kolk (2004). In a recent trial (Plascencia et al., unpublished), cattle fed a high-energy finishing diet supplemented with a mixture of ISQ-ALK (SANGROVIT-RS) increased flow of nonammonia N to small intestine, consequence of both decreased feed protein degradation and increased microbial protein synthesis (Table 11). SANGROVIT-RS supplementation did not affect VFA production, but decrease ruminal butyrate molar proportion (Table 12). Barajas et al. (unpublished) observed a 4% increase in dietary energy retention in finishing feedlot cattle supplemented with 4 g of SANGROVIT-RS/head daily.

Conclusions

Dynamics of ruminal fermentation account for a major portion of variation in feed efficiency and growth-performance of ruminants. A greater appreciation of the limits in rumen digestive function enables optimization of processes including protein synthesis, fiber digestion, and end-products of fermentation. Recent advances in technologies and development of new compounds has provided an assortment of tools than may be used to effectively modulate the fermentative processes.

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Table 3. Effects of urea level supplementation on digestive function and N metabolism in steers fed a 90:10 concentrate forage diet.

Item	Urea level			P - value		SEM
	1.25	0.7	0.1	Linear	Quadratic	
Steer replications	6	6	6			
Intake (g/d)						
OM	3343	3359	3378			
NDF	453	455	457			
N	66.2	57.6	48.2			
GE (Mcal/d)	15.2	15.2	15.3			
Flow to the duodenum (g/d)						
Total N	76.4	68.8	64.4	0.04	0.70	4.0
Microbial N	40.6	34.7	31.4	0.04	0.69	3.0
NH-N	2.30	2.04	1.58	0.06	0.72	0.3
Non-ammonia N	74.1	66.7	62.9	0.04	0.66	3.8
Feed N	24.7	23.2	22.7	0.20	0.71	0.9
Ruminal digestibility, %						
OM	62.6	58.3	53.23	0.04	0.91	0.3
NDF	25.3	18.6	9.30	0.05	0.54	0.9
Feed N	62.6	59.7	52.9	0.04	0.64	0.5
Microbial efficiency ¹	19.4	17.8	17.5	0.06	0.40	0.7
N efficiency ²	1.12	1.16	1.30	0.05	0.44	0.07
Postruminal digestibility (% of flow to duodenum)						
OM	62.2	59.7	60.0	0.37	0.49	1.9
NDF	15.3	11.2	22.8	0.41	0.33	7.3
Total N	72.6	67.6	66.2	0.08	0.53	2.7
Total tract digestibility (% of intake)						
OM	81.3	79.0	77.6	0.06	0.75	1.4
NDF	37.8	29.8	23.9	0.07	0.85	5.3
Total N	68.5	61.5	54.9	0.04	0.97	4.5
DE, %	78.5	76.0	74.5	0.05	0.74	1.4
DE, Mcal/kg	3.36	3.26	3.20	0.05	0.73	0.06

Adapted from May et al. (2014)

Table 4. Influence of treatments on characteristics of ruminal and total tract digestion in Holstein steers fed a diet supplemented with conventional urea or with combination of slow-release urea and conventional urea

Item	Treatments ¹		SEM	U-4.5 vs. OPT-4.5
	U-4.5	OPT-4.5		
Intake, g/d				
DM	4664	4637	64	0.02
OM	4403	4387	61	0.08
Starch	1903	2084	27	<0.01
ADF	452	501	7	<0.01
N	93	102	1.7	<0.02
GE, Mcal/d	21.69	21.36	0.39	0.01
Ruminal digestion, %				
OM	54.23	55.29	0.62	0.27
Starch	77.98	78.00	0.62	0.99
ADF	26.37	26.74	1.97	0.90
Feed N	66.22	68.38	1.10	0.15
Duodenal flow of N, g/d	94.18	101	2.7	<0.01
Duodenal flow of MN, g/d	61.06	65.70	2.01	<0.01
MN efficiency ²	23.24	25.07	0.44	0.02
Duodenal flow of NAN, g/d	92.52	97.80	0.83	<0.01
N efficiency ³	0.99	0.96	0.01	0.09
Postruminal digestion, % leaving abomasum				
OM	65.84	63.28	1.28	0.21
Starch	94.29	94.78	0.34	0.35
ADF	22.47	14.33	2.40	0.07
N	75.55	77.64	4.9	0.02
Total-tract digestion, %				
DM	77.65	76.81	0.57	0.28
OM	78.92	78.59	0.64	0.70
Starch	98.74	98.85	0.08	0.40
ADF	42.99	37.37	0.80	<0.01
N	75.28	77.75	0.64	<0.01
DE, %	78.30	78.23	0.53	0.91
DE diet, Mcal/kg	3.64	3.60	0.02	0.27
Observed-to-expected DE	1.00	1.02	0.006	0.02

¹ U-4.5 = Diet formulated to supply 2.07 Mcal/kg of NE_m with a ratio of starch:FDA of 4.5, containing 0.80% of conventional urea as the sole source of NNP and OPT-4.5=Diet formulated to supply 2.07 Mcal/kg of NE_m with a ratio of starch:FDA of 4.5, containing 0.80% of conventional urea plus 1.00% of Optigen. ²Microbial efficiency is expressed as duodenal MN, g/kg OM fermented in the rumen. ³ NAN= non-ammonia N. ¹¹N efficiency is expressed as duodenal non-ammonia N, g/g N intake. (López-Soto et al., 2014)

Table 5. Influence of treatments on characteristics of growth performance and dietary energy of diets in feedlot steers fed a diet supplemented with conventional urea or with combination of slow-release urea and conventional urea

Item	Treatments ^a				SEM
	SRU+U-4.5 ^c	U-3.5	U-4.5	U-5.5	
Pen replicates	5	5	5	5	
Number of steers	15	15	15	15	
Days on feed	70	70	70	70	
Weight, kg					
Initial	371.6	372.2	372.7	372.9	3.3
Final	461.2	453.3	462.1	473.9	6.5
Average daily gain, kg	1.279	1.158	1.276	1.443	0.052
Dry matter intake, kg	7.896	8.321	8.548	9.085	0.219
Dry matter intake, % LW	1.896	2.012	2.046	2.145	0.037
Gain for feed, kg/kg	0.162	0.139	0.149	0.159	0.003
Dietary net energy, Mcal/kg					
Maintenance	2.14	1.93	2.00	2.05	0.018
Gain	1.47	1.28	1.35	1.39	0.016
Observed to expected dietary ratio					
Maintenance	1.06	0.98	0.99	0.98	0.009
Gain	1.06	0.96	0.98	0.97	0.011
Observe to expected daily DM intake	0.94	1.02	1.02	1.04	0.011

^a SRU+U-4.5=0.80% conventional urea plus 0.80% of slow-release urea (Optigen II, Alltech Inc) at 4.5 S:F ratio, U-3.5 = 0.80 U at 3.5 S:F ratio, U-4.50=.80% U at 4.5 S:F ratio, and U-5.5= 0.80% U at 5.5 S:F ratio.
(López-Soto et al., 2014)

Table 6. Stratified meta-analysis of monensin effects on feed efficiency in growing and finishing cattle, across 4 decades of research (1970s to 2000s)

Decade, year	No. trials	Mean monensin dose, mg/kg feed	Weighted mean difference for feed efficiency, kg feed/kg BW gain	Control mean feed efficiency, kg feed/kg BW gain	Change in feed efficiency, %
1970s	85	26	-0.715	8.79	+ 8.1
1980s	21	31.7	-0.539	8.39	+6.4
1990s	13	33.9	-0.148	6.39	+2.3
2000+	11	32.3	-0.229	6.38	+3.5

Duffield et al. (2012)

Table 7. Effect of DL-malic acid supplementation on dry matter intake, milk yield and milk composition of Pelibuey ewes.

Item	Treatments ^a			SEM	Linear	Quadratic
	MA ₀	MA ₂	MA ₄			
DMI, g/wk	13,120	13,137	13,207	999	0.95	0.98
Milk yield, mL/wk	9,764	11,269	12,970	1,040	0.04	0.94
Milk efficiency ^b	0.74	0.86	0.98	0.07	0.03	0.82
Milk composition, %						
Protein	5.7	5.5	5.25	0.24	0.24	0.79
Fat	5.2	6.5	5.02	0.58	0.82	0.08
Total solids	15.0	16.3	15.78	0.76	0.46	0.36
Organic matter	94.8	94.8	94.69	0.32	0.75	0.93
Milk energy value, Kcal/g ^c	286	295	281	20.46	0.46	0.36
Milk components yield, g/wk						
Protein	557	620	681	49	0.02	0.62
Fat	508	732	651	70	0.14	0.11
Milk energy, Mcal/wk	2,792	3,324	3,645	213	0.01	0.48
FCM ^d	9,053	11,785	11,813	1.47	0.14	0.12
Milk efficiency (adjusted to FCM)	0.69	0.89	0.89	0.12	0.25	0.11
Performance of goat kids						
Average daily gain, g/d						
1 to 14 d	188	200	219	18.8	0.24	0.87
14 to 28 d	155	189	246	18.0	<0.01	0.62
1 to 28 d	170	195	232	14.1	<0.01	0.74

^a MA₀=No malic acid supplementation, MA₂= 2 g of MA/ kg of feed, and MA₄=4 g of MA/ kg of feed.

Adapted from Martínez-González et al. (2014)

Table 8. Effect of yeast supplementation on ruminal pH and digestion in cannulated Holstein steers (265 kg BW)

Item	Control	Live yeast level, g/ steer/d			SEM	Contrast <i>P</i> -value		
		10	20	30		Linear	Quadratic	Cubic
Ruminal pH ^b	6.19	6.25	6.08	6.38	0.125	0.49	0.37	0.26
Intake, g/d								
Dry matter	5,783	5,756	5,764	5,784	14	0.88	0.14	0.73
Organic matter	5,359	5,333	5,341	5,360	13	0.88	0.14	0.73
Neutral detergent fiber	970	965	966	970	2	0.88	0.14	0.73
Nitrogen	109	109	110	110	0.3	0.88	0.14	0.73
Gross energy, MJ/d	104.6	104.2	104.6	104.2	0.17	0.88	0.14	0.73
Ruminal digestion, %								
Organic matter	57.3	57.8	59.1	59.3	1.92	0.44	0.94	0.85
Neutral detergent fiber	32.2	41.3	41.1	43.8	2.66	0.03	0.28	0.34
Feed N	60.5	62.9	60.9	63.4	2.73	0.60	0.99	0.50
Microbial efficiency ^c	22.9	22.7	23.6	23.5	1.07	0.59	0.98	0.67
Protein efficiency ^d	0.979	0.979	0.982	0.978	0.028	0.74	0.67	0.69
Postruminal digestion, %								
Organic matter	65.1	65.0	67.1	67.7	1.87	0.26	0.70	0.63
Neutral detergent fiber	18.9	6.7	14.4	10.0	5.9	0.49	0.53	0.27
Nitrogen	68.3	67.9	68.2	68.3	2.35	0.97	0.91	0.92
Total tract digestion,%								
Dry matter	79.80	80.12	81.31	81.95	0.66	0.04	0.87	0.65

Organic matter	80.78	80.89	82.45	82.94	0.58	0.02	0.81	0.32
Neutral detergent fiber	45.3	45.7	49.8	49.8	1.26	0.02	0.91	0.22
Nitrogen	70.64	71.55	71.68	71.56	1.82	0.78	0.99	0.78
Digestible energy,%	78.84	79.16	80.96	81.0	0.46	0.05	0.95	0.39
Digestible energy, MJ/kg	14.27	14.31	14.68	14.69	0.065	0.05	0.96	0.39

^a Source of live yeast culture used was: *Saccharomyces cerevisiae* N. strain 7907, Biotecap®, Guadalajara, México, which contained 5.5×10^9 CFU/g (air-dry basis) and (g/g) 0.04 nitrogen, 0.06 NDF and 0.01 EE.

^b The ruminal pH was measured at 4-h posprandium (morning meal).

^c Microbial efficiency was expressed as: Duodenal microbial N, g/ kg of OM fermented in the rumen.

^d Protein efficiency was estimated as: Duodenal non-ammonia N, g/g of N intake.

(López-Soto et al., 2014)

Table 9. Treatment effects on growth performance and dietary energy in in feedlot steers fed different levels of a fibrolityc enzyme

Item	Fibrozyme level, g/d			
	0	5	10	SEM
Days on test	261	261	261	
Pen replicates	6	6	6	
Live weight (kg)				
Initial	182	186	183	4.7
Final ^a	499	504	532	5.8
Daily gain (kg) ^a	1.22	1.22	1.33	0.04
Dry matter intake (g/d)	5.94	6.12	6.28	0.11
Gain to feed (kg/kg)	0.205	0.200	0.212	0.003
Observed dietary net energy (Mcal/kg)				
Maintenance	2.37	2.29	2.39	0.04
Gain	1.67	1.60	1.69	0.04
Observed to expected dietary NE ^e ratio				
Maintenance	1.06	1.02	1.06	0.02
Gain	1.04	1.00	1.05	1.05
Carcass weight	321	324	342	6.4
Dressing percentage	63.7	64.7	65.00	0.6

^a Linear effect of supplementation level of Fibrozyme (P<0.05).

Adapted from Ware et al. (2002)

Table 10. Main effects of addition of condensed or hydrolysable tannins on 84-d feedlot growth performance of Holstein steers.

Item	Tannin treatments (g/h/d)				SEM	Effects		
	None	Queb	ChestN	Blend		Tannin	Que vs Che	Pure vs Blend
Live weight, kg								
Initial	392	391	393	391	2.2	0.91	0.48	0.67
Final	520	528	528	529	3.72	0.08	0.99	0.77
ADG, kg	1.53	1.64	1.61	1.65	0.05	0.08	0.67	0.60
DMI, g/d	10.0a	10.3ab	10.2ab	10.7b	0.15	0.06	0.66	0.05
ADG/DMI, kg/kg	0.152	0.159	0.157	0.155	0.01	0.42	0.84	0.60
Dietary NE, Mcal/kg								
Maintenance	2.12	2.17	2.17	2.11	0.04	0.59	1.00	0.39
Gain	1.45	1.49	1.49	1.45	0.04	0.59	1.00	0.39
Observe to expected dietary NE ratio								
Maintenance	0.96	0.98	0.98	0.95	0.02	0.59	1.00	0.39
Gain	0.94	0.97	0.97	0.94	0.02	0.59	1.00	0.39

Adapted from Rivera-Mendez et al.2014

Table 11. Influence of supplementation level of Sangrovit® RS on characteristics of ruminal and total tract digestion in cannulated Holstein steers (253.4 kg BW).

Item	Sangrovit® RS, g/head/day				SEM	Contrast <i>P</i> -value		
	0	2	4	6		Linear	Quadratic	Cubic
Intake (g/d)								
N	124	121	121	123	1.41	0.57	0.16	0.97
GE (Mcal/d)	23.95	23.44	23.36	23.73	0.27	0.57	0.16	0.97
Flow to duodenum (g/d)								
N	124	124	129	131	2.21	0.04	0.73	0.42
N-NH	3.95 ^a	3.27 ^b	2.46 ^c	2.55 ^c	0.13	<0.01	0.02	0.12
Nonammonia N	119.8 ^a	120.7 ^{ab}	126.7 ^{bc}	128.3 ^c	2.12	0.02	0.86	0.35
MN	62.18	62.17	65.01	64.67	1.31	0.13	0.91	0.34
Feed N	57.66	58.44	61.68	63.3	2.41	0.09	0.82	0.74
Ruminal digestion (%)								
Feed N	53.13	51.78	48.96	48.04	1.92	0.08	0.73	0.65
Microbial efficiency ¹	19.37 ^a	19.95 ^a	21.58 ^b	20.36 ^{ab}	0.40	0.04	0.11	0.11
N efficiency ²	0.97 ^a	1.00 ^a	1.05 ^b	1.05 ^b	0.02	0.04	0.59	0.45
Ruminal ammonia, mg/dL ³	17.1 ^a	13.1 ^{ab}	10.9 ^b	11.7 ^b	0.57	<0.01	0.05	0.84
Postruminal digestion leaving abomasum (%)								
N	71.13 ^a	73.21 ^a	76.28 ^b	75.79 ^b	0.68	<0.01	0.11	0.19
Total-tract digestion (%)								
N	71.18 ^a	72.64 ^{ab}	74.60 ^b	74.17 ^b	0.62	<0.01	0.16	0.35
DE (%)	78.25	78.40	80.31	79.59	0.68	0.09	0.54	0.20
DE diet (Mcal/kg)	3.48	3.48	3.57	3.54	0.03	0.09	0.54	0.20

Rows with different literals are different ($P < 0.05$).

¹Microbial efficiency is estimated as duodenal MN, g kg⁻¹ OM fermented in the rumen

²N efficiency is estimated as duodenal non-ammonia N, g g⁻¹ N intake.

³Ruminal ammonia values represent an average of ruminal concentrations determined at 4 and 8 hours post-feeding.

(Plascencia et al. unpublished)

Table 12. Influence of supplementation level of Sangrovit® RS on ruminal N-NH, pH, VFA concentration and estimated methane production

Item	Sangrovit® RS, g/head/day					Contrast <i>P</i> -value		
	0	2	4	6	SEM	Linear	Quadratic	Cubic
Time post-feeding = 4 hours								
pH	6.47	6.49	6.43	6.74	0.09	0.12	0.16	0.32
Ruminal ammonia, mg/dL	24.8 ^a	19.2 ^b	16.2 ^b	17.9 ^b	1.46	0.02	0.05	0.78
Total VFA, moles	80.2	78.3	77.1	75.3	4.54	0.46	0.99	0.95
Ruminal VFA, mol/100 mol								
Acetate	49.7 ^a	53.6 ^{ab}	53.5 ^{ab}	54.7 ^b	1.29	0.04	0.35	0.39
Propionate	31.2	33.4	31.9	31.3	2.76	0.91	0.66	0.69
Isobutyrate	0.23	0.13	0.40	0.26	0.10	0.50	0.17	0.08
Butyrate	14.3	9.9	11.4	11.1	1.98	0.37	0.32	0.41
Isovalerate	2.1	1.7	1.6	1.7	0.28	0.38	0.47	0.84
Valerate	2.9 ^a	1.5 ^b	1.5 ^b	1.2 ^b	0.25	<0.01	0.07	0.22
Acetate:propionate ratio	1.59	1.60	1.68	1.75	0.22	0.89	0.76	0.91
Methane production	0.44	0.44	0.45	0.46	0.03	0.33	0.70	0.33

Plascencia et al. unpublished)